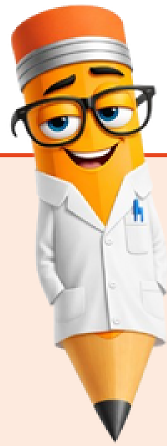
ALGEBRA & GRAPHS

Equations
& Changing the Subject

Linear, simultaneous, quadratic, fractional equations and formula rearrangement.



Use these notes well and you will be able to:

- Constructand solvelinear equationsinoneunknown.
- Solve fractional equations with algebraic denominators (F 02).
- Solve simultaneous linear equations (elimination and substitution).
- Solve one linear + one non-linear simultaneous equations (F 02).
- Solve quadratic equations: factorisation, completing the square, formula.
- Change the subject of a formula.

If you follow the steps, you will succeed.

Key Words

KEY WORD	SIMPLE DEFINITION
Linear equation	Highest power is 1. e.g. $3x + 4 = 10$. Straight-line graph.
Quadratic equation	Highest power is 2. e.g. $x^2 - 5x + 6 = 0$. Has up to 2 solutions.
Simultaneous equations	Two equations with two unknowns — solved together.
Elimination	Add or subtract equations to remove one unknown.
Substitution	Rearrange one equation, substitute into the other.
Completing the square	Writing x^2+bx+c as $(x+p)^2+q$ to find roots or vertex.
Quadratic formula	$x = (-b \pm \sqrt{(b^2-4ac)}) \div 2a$ — solves any quadratic.
Discriminant	b^2-4ac. >0: two roots. $=0$: one root. <0: no real roots.
Subject of formula	The variable on its own on one side of the formula.
Inverse operation	The opposite operation used when rearranging. ÷ undoes ×.

Solving Linear Equations

Goal: isolate the unknown. Apply INVERSE operations to both sides equally. Always: expand brackets first, collect like terms, then isolate.

Solve: $5 - 2x = 3(x + 7)$

Expand RHS: $5 - 2x = 3x + 21$
Collect x: $5 - 21 = 3x + 2x$
Simplify: $-16 = 5x$
$x = -16/5 = -3.2$

Solve fractional equation: $x/(x+2) = 3/(x-6)$

Cross-multiply: $x(x-6) = 3(x+2)$
Expand: $x^2 - 6x = 3x + 6$
Rearrange to zero: $x^2 - 9x - 6 = 0$
Use quadratic formula: $a=1$, $b=-9$, $c=-6$
$x = (9 \pm \sqrt{(81+24)}) / 2 = (9 \pm \sqrt{105}) / 2$
$x \approx 9.62$ or $x \approx -0.62$

Solve: $2/(x+2) + 3/(2x-1) = 1$

LCD = $(x+2)(2x-1)$. Multiply through:
$2(2x-1) + 3(x+2) = (x+2)(2x-1)$
$4x-2 + 3x+6 = 2x^2+3x-2$
$7x+4 = 2x^2+3x-2$ $2x^2-4x-6 = 0$ $x^2-2x-3 = 0$
$(x-3)(x+1) = 0$
$x=3$ or $x = -1$

Simultaneous Equations — Linear

Elimination method

Equations: $2x + 3y = 16$ ...**(1)**
$4x - y = 10$ ...**(2)**
Multiply (2) by 3: $12x - 3y = 30$ ...**(3)**
Add **(1)**+**(3)**: $14x = 46$ $x = 23/7$
Sub into **(1)**: $2(23/7)+3y=16$ $3y=16-46/7$
$3y= 66/7$ $y = 22/7$
$x = 23/7$, $y = 22/7$

Substitution method

Equations: $y = 3x - 1$ ...**(1)**
$2x + y = 9$ ...**(2)**
Sub **(1)** into **(2)**:
$2x + (3x-1) = 9$
$5x - 1 = 9$ $5x = 10$ $x = 2$
$y = 3(2)-1 = 5$
$x = 2$, $y = 5$

Simultaneous: One Linear, One Non-Linear

Solve: $y = x + 1$ and $x^2 + y^2 = 13$

Substitute $y = x+1$ into circle equation:
$x^2 + (x+1)^2 = 13$ $x^2 + x^2+2x+1 = 13$
$2x^2+2x-12 = 0$ $x^2+x-6 = 0$
$(x+3)(x-2) = 0$ $x = -3$ or $x = 2$
$y = (-3)+1 = -2$ or $y = 2+1 = 3$
Solutions: $(-3, -2)$ and $(2, 3)$

Solving Quadratic Equations

Method 1: Factorisation

Solve: $x^2 - 7x + 12 = 0$

Find two numbers: product = 12, sum = −7 −3 and −4
$(x - 3)(x - 4) = 0$
$x = 3$ or $x = 4$

Method 2: Completing the Square

Solve: $x^2 + 6x - 7 = 0$ by completing the square

Write: $(x + 3)^2 - 9 - 7 = 0$ $(x + 3)^2 = 16$
$x + 3 = \pm 4$
$x = 1$ or $x = -7$

Solve: $2x^2 - 4x - 3 = 0$ by completing the square

Divide by 2: $x^2 - 2x - 3/2 = 0$
$(x - 1)^2 - 1 - 3/2 = 0$ $(x-1)^2 = 5/2$
$x - 1 = \pm\sqrt{(5/2)} = \pm\sqrt{10}/2$
$x = 1 \pm \sqrt{10}/2$ (surd form)

Method 3: Quadratic Formula

QUADRATIC FORMULA

$$x = (-b \pm \sqrt{(b^2 - 4ac)}) \div 2a$$

$ax^2 + bx + c = 0$ | Discriminant = b^2-4ac: >0 two roots, =0 one root, <0 no real roots

Solve: $3x^2 + 5x - 2 = 0$ using the formula

$a=3$, $b=5$, $c=-2$. Discriminant = 25+24 = 49 $\sqrt{49} = 7$
$x = (-5 \pm 7) / 6$
$x = 2/6 = 1/3$ or $x=-12/6=-2$

Changing the Subject of a Formula

Apply inverse operations to isolate the required variable. If the subject appears twice — factorise it out first.

Make r the subject of: $A = \pi r^2$

Divide both sides by π: $A/\pi = r^2$
Square root both sides: $r = \sqrt{(A/\pi)}$
$r = \sqrt{(A/\pi)}$

Make x the subject of: $y = (x + 3) / (2x - 1)$

Multiply both sides by $(2x-1)$: $y(2x-1) = x+3$
Expand: $2xy - y = x + 3$
Collect x terms: $2xy - x = y + 3$
Factorise x: $x(2y - 1) = y + 3$
$x = (y + 3) / (2y - 1)$

ADD YOUR NOTES HERE

Hey smart student!

Withsimultaneousequations: substitution when one equation gives y= easily, elimination when coefficients can be matched. Pick the faster method.

Quadratic formula ALWAYS works — use it when factorising seems hard.

Keep going — see you in the next lesson! From FREDi